

about this book

This is not a book about Scala. This book is an introduction to *functional programming* (FP), a radical, principled approach to writing software. We use Scala as the vehicle to get there, but you can apply the lessons herein to programming in any language. As you work through this book, our goal is for you to gain a firm grasp of functional programming concepts, become comfortable writing purely functional programs, and be able to absorb new material on the subject, beyond what we cover here.

How this book is structured

The book is organized into four parts. In part 1, we talk about exactly what functional programming is and introduce some core concepts. The chapters in part 1 give an overview of fundamental techniques like how to organize small functional programs, define purely functional data structures, handle errors, and deal with state.

Building on this foundation, part 2 is a series of tutorials on *functional design*. We work through some examples of practical functional libraries, laying bare the thought process that goes into designing them.

While working through the libraries in part 2, it will become clear to you that these libraries follow certain patterns and contain some duplication. This will highlight the need for new and higher abstractions for writing more generalized libraries, and we introduce those abstractions in part 3. These are very powerful tools for reasoning about your code. Once you master them, they hold the promise of making you extraordinarily productive as a programmer.

Part 4 then bridges the remainder of the gap towards writing real-world applications that perform I/O (like working with databases, files, or video displays) and make use of mutable state, all in a purely functional way.

Throughout the book, we rely heavily on programming exercises, carefully sequenced to help you internalize the material. To understand functional programming, it's not enough to learn the theory abstractly. You have to fire up your text editor